

Route decision: Shanghai Global Distribution Center → Rotterdam Distripark

HUB-SHANGHAI → HUB-ROTTERDAM · mode any (STRICT) · cargo general · priority normal · scenarios: none · planned 2026-09-29 13:47 UTC · cargo value ¥786,575,545 at 25%/yr carrying cost

Recommendation: BALANCED, SEA via Malacca, Babel, Suez. Expected arrival in **580 h** (24.2 days); plan for 594 h (p85), worst case 614 h (p95). Cost: freight ¥613,332, landed ¥13,639,868 incl. ¥13,026,536 carrying cost. Peak leg risk 8%.

Options

Option	Mode	Via	Plan h	p50 h	p85 h	p95 h	Freight	Landed	Risk
FASTEST	AIR	direct	25	40	46	49	¥21,830,585	¥22,721,767	2%
SAFEST	AIR	direct	29	42	46	49	¥21,927,470	¥22,868,038	2%
BALANCED	SEA	Malacca, Babel, Suez	479	580	594	614	¥613,332	¥13,639,868	8%

ETA bands (bar p50–p95, tick p85, line plan → fully-correlated worst case)



Routes



Recommended route solid, alternatives dashed. Blue sea, amber air, green rail, orange road.

FASTEST: AIR via direct

AIR via no chokepoints; expected 39.7 h (p85 45.6 h, p95 49.3 h), ¥22,721,767 landed, 2 handoffs. 2.2 h faster than the next option. ¥9,081,959 more (landed) than BALANCED. Avoids Babel, Malacca, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	¥5,659	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	¥23,597	2%	
3	AIR	Frankfurt Airport	8,858	11.1	+8.8	¥20,902,476	0%	
4	AIR	Amsterdam Airport Cargo	366	0.5	+8.8	¥863,169	0%	
5	handoff	Amsterdam Airport Cargo		6.0	+0.0	¥23,597	2%	
6	ROAD	Rotterdam Distripark	64	0.9	+3.6	¥12,087	0%	

Transit 12.9 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 24.9 h. Model delay buffer p50/p85/p95: +14.7 / +24.9 / +32.0 h. Route band p50 39.7 h, p85 45.6 h, p95 49.3 h (independent legs); 56.9 h if all legs are late together. Costs: linehaul ¥21,783,390, transfers ¥47,195, disruption surcharge ¥0. Inventory carrying cost ¥891,182 expected (¥1,106,683 at p95); landed ¥22,721,767.

Why the slowest leg (AIR-SHANGHAI → AIR-FRANKFURT, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



SAFEST: AIR via direct

AIR via no chokepoints; expected 41.9 h (p85 46.3 h, p95 49.3 h), ¥22,868,038 landed, 2 handoffs. 2.2 h slower than FASTEST. ¥9,228,104 more (landed) than BALANCED. Avoids Babel, Malacca, Suez.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Shanghai Pudong International	30	0.4	+3.6	¥5,659	0%	
2	handoff	Shanghai Pudong International		6.0	+0.0	¥23,597	2%	
3	AIR	London Heathrow Airport	9,241	11.6	+8.8	¥21,806,622	0%	
4	handoff	London Heathrow Airport		6.0	+0.0	¥23,597	2%	
5	ROAD	Calais Channel Tunnel Hub	169	2.5	+3.6	¥31,929	0%	
6	ROAD	Rotterdam Distripark	191	2.8	+3.6	¥36,066	0%	

Transit 17.3 h + transfers 12.0 h + announced scenario delay 0.0 h = plan 29.3 h. Model delay buffer p50/p85/p95: +12.6 / +19.8 / +25.4 h. Route band p50 41.9 h, p85 46.3 h, p95 49.3 h (independent legs); 54.6 h if all legs are late together. Costs: linehaul ¥21,880,276, transfers ¥47,195, disruption surcharge ¥0. Inventory carrying cost ¥940,568 expected (¥1,106,683 at p95); landed ¥22,868,038.

Why the slowest leg (AIR-SHANGHAI → AIR-HEATHROW, p85 8.8 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



BALANCED: SEA via Malacca, Babel, Suez

SEA via Malacca, Babel, Suez; expected 580.3 h (p85 594.3 h, p95 613.8 h), ¥13,639,868 landed, 3 handoffs. 540.6 h slower than FASTEST. ¥9,081,959 cheaper (landed) than the lowest-cost alternative.

#	Mode	To	km	Hours	Delay p85	Cost	Risk	Signal
1	ROAD	Port of Shanghai	4	0.1	+3.6	¥794	0%	
2	handoff	Port of Shanghai		14.0	+0.0	¥39,329	8%	
3	SEA	Strait of Malacca	3,810	108.9	+26.6	¥89,910	0%	
4	SEA	Bab el-Mandeb Strait	6,585	188.1	+27.5	¥155,385	0%	
5	SEA	Suez Canal	2,302	65.8	+28.5	¥54,326	0%	
6	SEA	Port of Genoa	2,552	72.9	+20.6	¥60,231	0%	
7	handoff	Port of Genoa		8.0	+0.0	¥28,317	5%	
8	RAIL	Lyon Rail Freight Hub	356	5.9	+15.6	¥27,978	0%	
9	handoff	Lyon Rail Freight Hub		4.0	+0.0	¥15,732	3%	
10	ROAD	Paris Charles de Gaulle	403	5.9	+3.6	¥76,122	0%	
11	ROAD	Lille Logistics Hub	184	2.7	+3.6	¥34,688	0%	
12	ROAD	Rotterdam Distripark	162	2.4	+3.6	¥30,524	0%	

Transit 452.7 h + transfers 26.0 h + announced scenario delay 0.0 h = plan 478.7 h. Model delay buffer p50/p85/p95: +101.6 / +133.4 / +179.3 h. Route band p50 580.3 h, p85 594.3 h, p95 613.8 h (independent legs); 658.0 h if all legs are late together. Costs: linehaul ¥529,955, transfers ¥83,377, disruption surcharge ¥0. Inventory carrying cost ¥13,026,536 expected (¥13,778,541 at p95); landed ¥13,639,868.

Why the slowest leg (CHOKE-BABEL → CHOKE-SUEZ, p85 28.5 h) is slow: exact Shapley attribution of the p85 delay model, hours relative to a 28.9 h baseline



Method: legs are priced with gradient-boosted quantile models (p50/p85/p95) trained on synthetic delays fitted to published UNCTAD, World Bank and STB statistics; FASTEST plans on p50, BALANCED on p85 (plus cargo carrying cost when a value is given), SAFEST on p95 plus 240 risk-hours per unit of threat. Route bands assume independent legs. Sea distances are great-circle between chokepoint waypoints. Currency conversion: live (open.er-api.com) as of Tue, 29 Sep 2026 00:02:31 +0000.